

Interpreting CNN in Cybersecurity: Process-Level Attribution via Benign-Replacement Occlusion (BRO)

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Abstract—Convolutional Neural Networks (CNN) have shown strong performance in classifying malware-infected virtual machines (VM) by analyzing snapshots of running processes and their associated metrics. While prior work has explored methods for interpreting these decisions, gradient-based approaches often lack the precision necessary to clearly reveal which features the model relies on during decision-making. To address this, we employed a saliency-based occlusion method that masks individual input elements to estimate their importance. We propose a novel benign-replacement occlusion (BRO) technique, which replaces each feature with its counterpart from a known benign sample. By preserving the statistical structure of the input, BRO yields a more faithful representation of feature importance. Our results demonstrate that BRO enhances CNN interpretability in cybersecurity contexts—particularly in identifying what processes/metrics determine an infected system.

Index Terms—Convolutional neural networks (CNN), deep learning, malware detection, occlusion, saliency masking, security, visualizations.

I. INTRODUCTION

Cybersecurity has grown increasingly complex as modern malware adopts stealthier techniques to evade detection. Traditional signature-based approaches often fail to identify new or obscure threats, motivating a shift toward data-driven methods. Convolutional Neural Networks (CNN), in particular, have shown remarkable accuracy in detecting malware infections from structured snapshots of running processes and system metrics in virtual machines (VM). However, high classification performance alone is insufficient for real-world applications—security analysts must also understand why a model made a decision, especially to isolate and mitigate the offending processes.

This work addresses the need for reliable interpretability in CNN-based malware detection. We present a feature attribution technique based on saliency-driven occlusion, which quantifies the importance of individual process-metric pairs by measuring the change in model confidence when they are masked. Our method is designed to help analysts trace predictions back to specific system activity, enhancing trust and enabling actionable insights.

We compare two occlusion strategies: (1) zero replacement, a common approach that masks features by setting them to

zero, and (2) a novel benign replacement occlusion (BRO) method, which substitutes each masked value with its counterpart from a benign (non-infected) sample. While zeroing introduces stark disruptions that may not reflect natural process behavior, BRO retains realistic structure in the input data, helping to avoid misleading attribution caused by unnatural perturbations.

Across 320 controlled experiments—spanning four CNN architectures, two input orderings [1], and 40 labeled VM samples—we observe that BRO reveals sharper drops in confidence for key features in malignant cases, especially for known processes like `systemd-logind` and metrics like `mem_rss`. Our results suggest that BRO may offer a more precise, class-aware understanding of feature relevance than zeroing alone.

This paper makes the following contributions:

A. Occlusion-based feature attribution

We introduce a method that attributes CNN decisions to individual processes and metrics by systematically masking them and observing the resulting confidence drop.

B. Evaluation of zero vs. BRO

We compare zero masking with a novel BRO approach across multiple CNN architectures and ordering schemes, revealing model- and class-specific patterns in attribution behavior.

C. Practical visualization tools

We provide per-sample heat maps and ranked drop charts that help analysts visualize which features most influence model decisions, offering interpretable outputs for cybersecurity scenarios.

The remainder of this paper is organized as follows: Section II reviews related work; Section III details our experimental setup; Section IV presents our findings; and Section V concludes with future directions.

II. RELATED WORKS

CNNs are increasingly and successfully used for malware detection in virtual machines by analyzing process-level metrics [2], [1], though their decision-making remains difficult to interpret. MiCAM [3] addressed this by applying gradient-based visualizations to structured data; we build on this direction with an occlusion-based approach that remains effective across different input orderings. Using the dataset from Abdelsalam et al.[2], which captures benign and malicious VM behavior at 10-second intervals, we extend interpretability efforts into the cybersecurity domain. Prior work has used occlusion to probe or improve CNNs[4], [5], [6], [7]; Osharov and Lindenbaum [4] applied benign-like perturbations to natural images, while our method adapts this concept to structured process data to reveal which features most influence model confidence.

III. METHODOLOGY

A. Data Acquisition and Preparation

Our dataset consists of structured TFRecord files derived from the experimental setup in [2], where labeled VM snapshots were collected at 10-second intervals during controlled malware injections. Each snapshot is formatted as a 2D numerical array, with rows representing active system processes and columns representing per-process metrics (e.g., `mem_rss`, `cpu_percent`). Samples are labeled as benign (0) or malignant (1) depending on whether they were collected before or after the introduction of malware.

Following prior work on feature arrangement [1], each sample was evaluated under two distinct orderings: a low-performing baseline and a high-performing arrangement shown to improve CNN classification accuracy.

B. CNN Model Development

We trained four different CNN architectures: LeNet5, Xception, InceptionV3, and DenseNet121. Each model was trained to classify TFRecord samples as benign or malignant based on their process-metric matrix. Model-specific training schedules and hyperparameters were tuned to balance accuracy and generalization. To ensure fair comparison across interpretability methods, all occlusion-based evaluations used the same set of trained models and test samples.

C. Visualization Techniques

We initially employed MiCAM [3], a Grad-CAM adaptation for structured data, to interpret CNN predictions. However, MiCAM consistently failed to isolate meaningful individual features. Its heatmaps often showed uniform or diffuse activations and reflected gradient sensitivity rather than concrete attribution. Since MiCAM was unable to reliably identify specific processes or metrics responsible for predictions, we pursued an occlusion-based strategy better suited to our structured data format.

D. Occlusion-Based Interpretability

We employed two occlusion strategies to assess the importance of individual features:

- **Zero-replacement:** each process, metric, or cell is masked by replacing its value(s) with 0.
- **Benign-replacement:** each feature is replaced with the average value computed across benign samples at the same row and column index.

Our occlusion approach draws inspiration from prior work in model interpretability, where feature masking has been used to assess saliency by measuring changes in prediction confidence [8], [6]. In these methods, important features are identified by the degree to which masking them impacts the model’s output.

The BRO strategy was designed to address a limitation in the zero-replacement approach, where inserting unnatural zeros could exaggerate the perceived importance of a feature by distorting the input distribution. Instead, BRO preserves a plausible input by drawing realistic values from the benign class.

Each masking method was applied at three granularities:

- **Row-level masking:** entire process rows were replaced.
- **Column-level masking:** entire metric columns were replaced.
- **Cell-level masking:** individual process-metric entries were replaced.

After masking, the altered sample was re-evaluated by the model, and the resulting prediction confidence was compared to the original. Specifically, we computed the drop in softmax confidence for the initially predicted class:

$$\text{Drop} = p_{\text{original}} - p_{\text{masked}}$$

This confidence drop was treated as an estimate of the feature’s contribution to the model’s decision. A larger drop implied greater importance.

E. Experimental Setup

To enable side-by-side comparison, all experiments were conducted in a unified pipeline. For each sample, both occlusion strategies were applied using the same trained model and prediction context. This ensured that any differences in attribution could be attributed solely to the masking method, not to differences in training or data.

We evaluated 40 test samples under both ordering schemes across all four CNN architectures, resulting in 320 model-input combinations per occlusion method. For each sample, we masked and evaluated every process row, every metric column, and every individual cell.

All computed drops were recorded into a structured database, allowing for later aggregation and filtering. Row and column results were displayed as bar graphs, while cell-level results were visualized using 2D heatmaps aligned with the process-metric layout of the input.

F. Data Splitting

Training and testing samples were selected to avoid direct overlap, ensuring that test-time evaluations were performed on data not seen during training. However, samples from the same VM instance may appear in both classes (e.g., before and after infection), so our analysis focuses on interpretability rather than full isolation across machine identity.

IV. RESULTS

A. Initial Hypothesis Failed

Initially, we hypothesized that interpretability could be derived from gradient-based mappings of pooling and convolutional layers [3]. However, as the experiments progressed, the resulting “hot spots” did not consistently correspond to specific processes or metrics.

This result is understandable in hindsight: gradient-based methods highlight areas of rapid local change, which may not correspond to the absolute importance of individual processes or metrics for interpretability.

The following figure, Fig. 1, shows a pooling layer heatmap that is the closest to the results we were getting with the occlusion-based method. Although the hot spots seem to be pinpointing some rows/processes and columns/metrics, the occlusion based method will show that the current hot spots have little effect on the accuracy of the model’s decision.

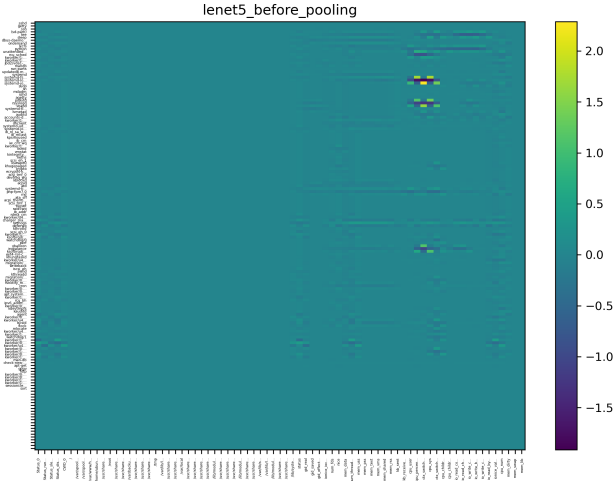


Fig. 1. Gradient-based map before pooling of sample 115 (99.99% malignant) using Lenet5 after 5-epochs. Rows/columns are processes/metrics as labeled.

B. Promising Results Through Occlusion

To overcome the limitations of gradient-based visualizations, we adopted saliency-based masking (occlusion), which has shown promise in interpreting CNNs for visual data [4], [5], [6], [7] but remains underused for structured, non-visual inputs. Applying zero-based occlusion to individual rows and columns in our samples revealed that `systemd-logind` and `mem_rss` consistently triggered the largest confidence drops—sometimes up to 98%—highlighting their importance

in classification. These results validated occlusion as a more effective method for identifying critical features in our context than gradient-based alternatives.

C. Visualizing the Results

Obtaining the former information prompted the development of a way to visualize our findings. There are now visualization tools that are provided through Keras and TensorFlow, however environmental restrictions demanded a custom tool tailored for the environment that can map the significance of occluding data. We present the visualization of occlusion at the cell level in Fig. 2.

Fig. 2, our occlusion-based heatmap, shows a spike in confidence drop for a singular row and column. The figure is one of the more polarized figures, but later in the paper, the discussion will present the average confidence drop for each process/metric in each model and data ordering. To show a better overview of the entire experiment.

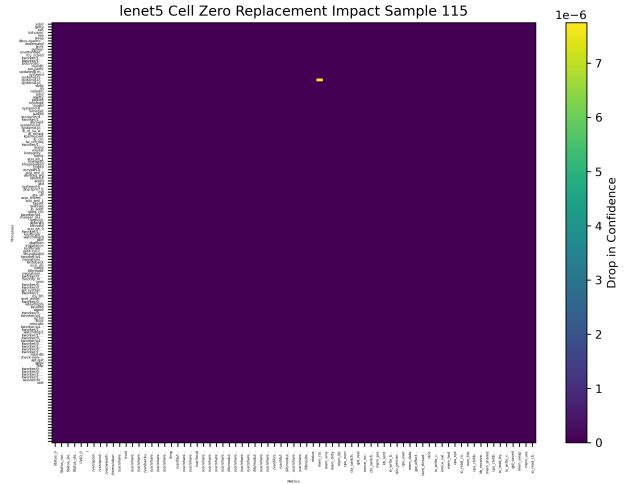


Fig. 2. occlusion-based map of sample 115 (100% malignant) using Lenet5 after 20-epochs. Rows/columns are processes/metrics as labeled.

D. Addressing the Converse

Interestingly, when the model was given a benign VM to classify, occlusion seemed to have little to no effect on the confidence of the model as shown in the following Fig. 3, The results from benign VM showed more deviation from the usual culprits, if they showed any drop in confidence at all. Because masking these processes/metrics had little to no effect on the overall accuracy, we decided to also show the average of drop per process/metric according to the class. We hypothesized that the benign class was deflating the overall drops significantly.

E. Feature Importance and Aggregated Rankings

To assess the reliability of zero-replacement occlusion, we applied it across four CNN architectures—LeNet5, Xception, DenseNet121, and InceptionV3—using both their worst- and best-performing input orderings based on prior research [1]. For each model, we measured the average drop in confidence

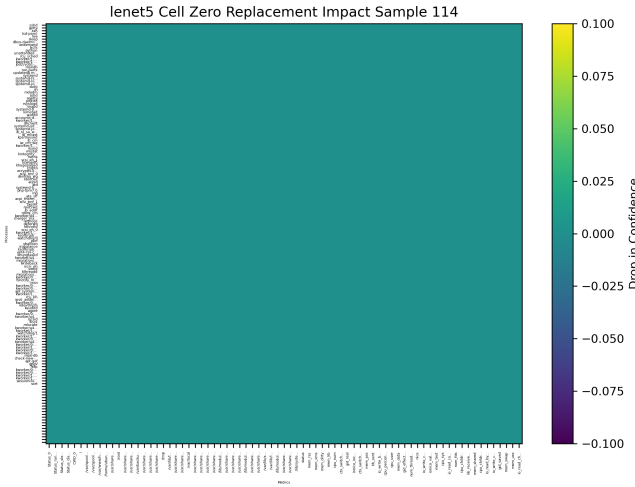


Fig. 3. Occlusion-based heatmap of sample 114 (100% benign) using Lenet5 after 20-epochs. Where each row is labeled by process and each column by metric

(avg_drop) when masking individual processes or metrics, ranking features by their influence. Consistent top contributors emerged across models, particularly `systemd-logind` and `mem_rss`, supporting zero-replacement’s ability to highlight key decision-driving elements regardless of architecture.

Table I presents the top five most impactful features for each model, and aggregates those results across all 320 experiments to highlight global trends.

TABLE I
TOP 5 MOST IMPORTANT ELEMENTS PER MODEL (BY AVG_DROP, ZERO REPLACEMENT)

Model	Element Type	Element Name	Avg Drop
Lenet5	process	systemd-logind	0.4361
	metric	mem_shared	0.0460
	metric	mem_rss	0.0329
	process	python	0.0171
	metric	mem_vms	0.0141
Xception	process	systemd-logind	0.3678
	metric	mem_rss	0.2130
	process	systemd-journald	0.1738
	process	kworker/u4:2	0.0316
	process	kworker/u4:0	0.0285
InceptionV3	metric	mem_vms	0.2489
	metric	status	0.2446
	process	systemd-logind	0.1821
	metric	mem_rss	0.1777
	process	tee	0.0984
Densenet121	process	systemd-logind	0.4519
	metric	mem_rss	0.2470
	metric	mem_shared	0.1464
	metric	mem_vms	0.1395
	metric	Status_0	0.1016
ALL	process	systemd-logind	0.3595
	metric	mem_rss	0.1676
	metric	mem_vms	0.1020
	metric	status	0.0791
	metric	mem_shared	0.0566

F. Introducing Benign Replacement Occlusion (BRO)

While zero-replacement occlusion proved effective in identifying impactful features, it raised a critical concern: introducing zeros into structured data creates unrealistic conditions. Since many processes and metrics rarely or never take zero values in real system states, this artificial substitution could distort the model’s behavior and lead to inflated importance scores.

To address this, we developed a second occlusion strategy: *BRO*. Instead of masking a feature with zero, we replace it with the corresponding average value from known benign samples. For example, when masking the process `systemd-logind`, we substitute its metric values with the average [row, column] values collected from benign TFRecords, respectively. This preserves the internal consistency of the input while still disrupting any potentially malicious signal associated with the feature.

The BRO method provides a more natural perturbation, keeping the input within the distribution of valid system behavior. It aims to reveal how much a given process or metric contributes to classification—while avoiding the unrealistic assumptions inherent in zero masking.

We applied this technique to the same set of 320 model-sample combinations across all four CNN architectures and both ordering conditions. The results are shown in Table II, which mirrors the format of our zero-replacement summary and highlights the top five most influential elements for each model. An aggregated block at the bottom summarizes the most impactful features across all models combined.

TABLE II
TOP 5 MOST IMPORTANT ELEMENTS PER MODEL (BY AVG_DROP, BRO)

Model	Element Type	Element Name	Avg Drop
Lenet5	process	systemd-logind	0.4235
	metric	mem_rss	0.1502
	process	python	0.1017
	metric	mem_shared	0.0755
	metric	mem_vms	0.0486
Xception	process	systemd-logind	0.3749
	metric	mem_rss	0.2504
	process	systemd-journald	0.1887
	process	(sd-pam)	0.1269
	metric	mem_vms	0.0540
InceptionV3	metric	mem_vms	0.2949
	metric	mem_rss	0.2712
	metric	mem_pss	0.2531
	metric	status	0.2478
	process	systemd-logind	0.1841
Densenet121	process	systemd-logind	0.3882
	metric	mem_vms	0.3227
	process	lxcfs	0.2546
	metric	mem_rss	0.2382
	metric	Status_0	0.1882
ALL	process	systemd-logind	0.3427
	metric	mem_rss	0.2275
	metric	mem_vms	0.1801
	metric	status	0.1003
	metric	mem_shared	0.0972

G. Comparing Zero and BRO Methods

While both occlusion strategies—zero replacement and BRO—consistently identified high-impact features, their outcomes diverged in ways that may reflect deeper differences in how they probe model behavior. Critical elements like `systemd-logind` and `mem_rss` appeared prominently in both methods across multiple architectures, but their measured importance sometimes flipped—suggesting that the two approaches may surface distinct attribution patterns.

In LeNet5, for instance, `systemd-logind` showed a slightly lower average drop under BRO (0.4235) compared to zero replacement (0.4361), but `mem_rss` produced a much higher average drop under benign (0.1502) than zero (0.0329). This suggests that BRO may be assigning greater importance to metrics like `mem_rss`, while reducing the apparent impact of other features. In other words, the model’s “saliency map” may shift under the two occlusion methods.

This trend is echoed across other models as well. In Xception, `mem_rss` rose from 0.2130 (zero) to 0.2504 (benign), and in DenseNet121, `mem_vms` jumped from 0.1395 to 0.3227. Such increases suggest that BRO, by maintaining realistic input distributions, may enhance signal clarity for certain metrics while suppressing artifacts introduced by unnatural masking.

Another notable difference is the emergence of new top-ranked features under benign masking—such as `lxcfs`, `Status_0`, and `mem_pss`—which were largely absent under zero masking. This supports the idea that BRO not only reveals similar core features but may also uncover additional decision pathways that the zero method fails to highlight.

However, the overall average drops for BRO are often lower than those of zero masking when looking across all samples. This can be misleading. Because benign masking replaces values with clean, expected data, it causes minimal disruption in benign samples. As a result, confidence drops for those samples are small—even when masking important features—leading to a deflation of average importance scores. This stands in contrast to zero replacement, which perturbs all samples equally, often exaggerating importance even when it’s not warranted.

This raises an important hypothesis: BRO may yield lower averages overall not because it’s less effective, but because it better preserves model confidence on clean data. If we isolate only malignant samples, we may discover that BRO produces even sharper and more informative confidence drops, particularly for malware-relevant features.

H. Class-Specific Drop Patterns

To further evaluate whether the BRO method improves feature attribution, we analyzed drop magnitudes separately for benign and malignant samples. Table III summarizes the class-wise average drop for `systemd-logind` and `mem_rss` across all models and both replacement strategies.

Several notable patterns emerge. For `mem_rss`, the BRO consistently produces higher average drops than zero replacement on malignant samples across all models. This

suggests that the BRO method better identifies metrics tied directly to the model’s confidence in classifying infections. For `systemd-logind`, BRO typically yields higher drops as well, with the exception of DenseNet121.

In contrast, for benign samples, the BRO leads to much smaller drops—frequently near zero or even negative. This aligns with expectations: masking benign data with benign-like values causes little disruption. Zero replacement, however, introduces unnatural distortions, resulting in larger shifts in confidence even when masking non-discriminative features.

These results suggest that BRO offers more reliable, class-aligned attribution—particularly for interpreting malware classifications. However, because benign samples cause minimal impact under this method, they can dilute overall averages when not separated by class. This reinforces the importance of class-aware attribution analysis.

Finally, the appearance of negative average drops—where masking improves model confidence—raises new questions about feature redundancy and interference.

TABLE III
CLASS-WISE AVG DROP COMPARISON FOR ZERO VS. BRO

Model	Element	Class	Zero Drops	Benign Drops
LeNet5	<code>systemd-logind</code>	1	0.5645	0.9421
		0	0.3311	-0.0008
	<code>mem_rss</code>	1	0.0317	0.3340
Xception	<code>systemd-logind</code>	0	0.0338	-0.0002
		1	0.8116	0.8284
	<code>mem_rss</code>	0	0.0047	0.0040
InceptionV3	<code>systemd-logind</code>	1	0.4302	0.5077
		0	0.0353	0.0399
	<code>mem_rss</code>	1	0.3904	0.3986
DenseNet121	<code>systemd-logind</code>	0	0.0118	0.0087
		1	0.3389	0.5054
	<code>mem_rss</code>	0	0.0458	0.0795
ALL	<code>systemd-logind</code>	1	0.8724	0.8689
		0	0.1078	-0.0051
	<code>mem_rss</code>	1	0.5224	0.5324
	<code>systemd-logind</code>	0	0.0216	-0.0025
		1	0.6597	0.7595
	<code>mem_rss</code>	0	0.1138	0.0017
	<code>systemd-logind</code>	1	0.3308	0.4699
		0	0.0341	0.0292
	<code>mem_rss</code>	1	0.3308	0.4699
		0	0.0341	0.0292

I. Negative-Drop Features

In rare cases, masking certain features increased the model’s confidence, indicating they may introduce noise. Under zero replacement, features like `systemd-fsckd`, `rcu_bh`, and `ib_mcast` showed small negative drops, with confidence increasing by up to 0.0015.

BRO revealed similar trends but with smaller magnitudes. For example, masking `systemd-fsckd` still improved confidence, though only marginally (about 0.0004), suggesting that BRO is less prone to overstating the negative influence of weak features.

While the effects are minimal, this reinforces the idea that benign masking may yield cleaner attributions, especially in noisy or low-signal cases.

J. Residual Accuracy and Redundancy

Summing the individual `avg_drop` values across all occluded features does not account for 100% of the model's accuracy. The residual performance—sometimes exceeding 40%—suggests the following:

- 1) **Redundancy:** Overlapping information exists across multiple processes and metrics.
- 2) **Nonlinear interactions:** Convolutional neural networks leverage complex combinations of features that go beyond single-feature effects.

Thus, while occlusion quantifies first-order importance, a complete understanding of the model's decision-making must consider higher-order interactions.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

This study demonstrated that saliency-based occlusion, particularly our novel BRO method, can effectively identify the most influential features driving CNN-based malware classification. Across 320 sample predictions using four architectures (LeNet-5, Xception, InceptionV3, DenseNet121), we consistently observed `systemd-logind` and `mem_rss` as top contributors to model confidence, with accuracy drops reaching over 75% in malignant samples when these were occluded.

We compared traditional zero-replacement with our BRO technique, which substitutes each feature with class-averaged benign values. Our analysis revealed that BRO often led to larger confidence drops for critical features in malignant samples, suggesting it more faithfully disrupts the signal learned for identifying infections. In contrast, zero replacement sometimes underestimated a feature's importance by introducing out-of-distribution inputs.

Architecture-wise rankings showed that despite variation in convolutional depth and receptive fields, all four CNNs converged on a narrow set of high-impact features. These features maintained their importance across different input orderings, confirming the robustness of the occlusion findings. Aggregated rankings across all models reaffirmed the dominance of a few key processes and metrics.

Class-wise analysis showed that benign samples produced consistently lower drops than malignant ones, with BRO appearing to suppress model confidence less aggressively for clean inputs. This suggests the method may preserve decision boundaries more faithfully, especially when interpreting positive detections. These findings motivated our split-by-class evaluations, which provided more nuanced insights into feature importance per class.

Although rare, a handful of negative-drop features emerged under both masking methods, subtly improving classification when occluded. While their impact was minor, they indicate areas of potential model confusion or data noise.

Overall, our framework offers a robust, interpretable lens into CNN decision-making, providing process- and metric-level insight that may inform real-world malware triage and defense.

B. Future Work

- 1) **Real-time Integration:** Integrate occlusion analysis into intrusion-detection pipelines to automatically flag influential features during live classification.
- 2) **Multi-Feature Interactions:** Extend masking strategies to jointly occlude feature pairs or groups, revealing non-linear synergies between elements.
- 3) **Adaptive Ordering:** Develop dynamic or learned feature ordering strategies to optimize CNN performance and interpretability on structured tabular inputs.
- 4) **Comparative Interpretability:** Benchmark occlusion methods against model-agnostic tools like SHAP and LIME to assess relative interpretability and efficiency.
- 5) **Application to Natural Data:** Explore the efficacy of our BRO occlusion technique on natural datasets (e.g., images, audio) to assess its generalizability beyond structured metrics.

By pursuing these directions, we aim to transform CNN interpretability from a post-hoc research exercise into a practical toolset for threat analysts, enabling precise, data-driven response to emerging cyber threats.

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